BIO303

MID TERM SUBJECTIVE BEST FILE BY

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Define oxidation and reduction reaction

Oxidation: Oxidation is defined as “addition of oxygen or any electronegative element and removal of hydrogen or any electropositive element”. Reduction: Reduction is defined as” addition of hydrogen or any electropositive element and removal of oxygen or any electronegative element”.

What is the Lyase.

an enzyme (such as a decarboxylase) that forms double bonds by removing groups from a substrate other than by hydrolysis or that adds groups to double bonds.

Write three form of GULT

- Class 1 (GLUTs 1–4, 14);
- Class 2 (GLUTs 5, 7, 9, and 11);
- Class 3 (GLUTs 6, 8, 10, 12, and HMIT)

Glucose 6phophate conversion:

glucose-1-phosphate is converted (reversibly) to glucose-6-phosphate by the enzyme phosphoglucomutase. Those tissues also house the enzyme glucose-6-phosphatase, which converts glucose-6-phosphate into free glucose that is secreted into the blood, thereby restoring blood glucose levels to normal.

Isozymes:

In biochemistry, isozymes (also known as isoenzymes or more generally as multiple forms of enzymes) are enzymes that differ in amino acid sequence but catalyze the same chemical reaction. Isozymes usually have different kinetic parameters (e.g. different KM values), or are regulated differently.

Kinase:

(KY-nays) A type of enzyme (a protein that speeds up chemical reactions in the body) that adds chemicals called phosphates to other molecules, such as sugars or proteins. This may cause other molecules in the cell to become either active or inactive.

Conversion of Glucose 6-Phosphate to Fructose 6-Phosphate

The second step of glycolysis involves the conversion of glucose-6-phosphate to fructose-6-phosphate (F6P). This reaction occurs with the help of the enzyme phosphoglucose isomerase (PI). As the name of the enzyme suggests, this reaction involves an isomerization reaction.

Standard reduction potential

The tendency for a chemical species to be reduced, **at standard conditions. The more positive the potential is the more likely it will be reduced…** denoted by E^{O}.

DNA and RNA

Difference between DNA and RNA are:

DNA	RNA
It is double stranded nucleic acid.	It is single stranded nucleic acid.
It contains deoxyribise sugar.	It contains ribose sugar.
It contains Thymine (T) as a nitrogenous base.	It contains Uracil (U) instead of Thymine.
It is the genetic and hereditary material of the cells.	It is involved in synthesis of proteins.
It is present in the nucleus of the cells.	It is present in both nucleus and cytoplasm.

How many nucleotides contain one molecule of tRNA.

t RNA molecules consist of 74-95 nucleotides in a particular sequence.

T phy c arm in ribose it's function:

All tRNA molecules contain 4 main arms or loops.

1-Acceptor arm:

This is made up of unpaired sequences of cytosine-cytosineadenine (CCA) at the 3′end.

2-Anticodon arm:

It is in the form of a loop and carries specific sequences of three bases which constitute the anticodon.

3-D arm:

It contains the base dihydrouridine.

4-TΨC arm:

It contains thymine, pseudouridine and cytosine.

Write about Elimination reaction:

In these reactions, two atoms or groups are removed from a molecule without being substituted by other atoms or groups. A new C-C double bond is formed between the carbon atoms to which the eliminated atoms or groups are previously attached. It is always accompanied with change in hybridization.

Hydrolases:

Hydrolase is a class of hydrolytic enzymes that are commonly used as biochemical catalysts which utilize water as a hydroxyl group donor during the substrate breakdown. In simple words, a hydrolase is an enzyme that catalyzes the hydrolysis of a chemical bond in biomolecules.

What kind of reaction ATP produce energy:

When one phosphate group is removed by breaking a phosphoanhydride bond in a process called hydrolysis, energy is released, and ATP is converted to adenosine diphosphate (ADP). Likewise, energy is also released when a phosphate is removed from ADP to form adenosine monophosphate (AMP).

Hydrolysis:

Hydrolysis is a common form of a chemical reaction where water is mostly used to break down the chemical bonds that exists between a particular substance. Hydrolysis is derived from a Greek word hydro meaning water and lysis which translates to the word break or to unbind.

Ping-pong mechanism of nucleoside diphosphate kinase

A ping-pong mechanism integrates phosphorylation of a histidine residue by transferring terminal phosphate group (γ-phosphate) from ATP to NDP β-phosphate in order to produce a NTP, and NDPK catalyzes such reversible reactions. NTP phosphorylates a histidine, which in turn phosphorylates NDP.

Isomerization:

Isomerisation is the chemical process by which one molecule is transformed into another molecule which has exactly the same atoms in same numbers, but the atoms have a different arrangement e.g. A−B−C→B−A−C (these related molecules are known as isomers). Chemistry.

Cis trans isomerism

Cis-trans isomers exhibit a type of stereoisomerism where the atoms have different spatial arrangements in three-dimensional space.

Alternative paths from pyruvate to phosphoenolpyruvate

In the first step of gluconeogenesis, pyruvate carboxylase (with ATP and bicarbonate) converts pyruvate to oxaloacetate. Then phosphoenolpyruvate carboxykinase (PEPCK) (with GTP) releases carbon dioxide to give phosphoenolpyruvate.

what are the 5 ion channel effect diseases:

Mutations in muscle voltage-gated sodium, potassium, calcium and chloride channels, and acetylcholine-gated channel may lead to such physiological disorders as hyper- and hypokalemic periodic paraly- sis, myotonias, long QT syndrome, Brugada syndrome, malignant hyperthermia and myasthenia.

what are the steps of the oxidative phase of ppp pathway:

This phase is made up of 2 irreversible steps:
Step 1:
Glucose-6-phosphate is oxidized to form lactone. NADPH is produced as a byproduct of this reaction as NADP+ start superscript, plus, end superscript is reduced as glucose-6-phosphate is oxidized.
Step 2:
Next, a carbon is removed (cleaved) and CO_2 is released. Once again, the electrons released from this cleavage is used to reduce NADP+ start superscript, plus, end superscript to NADPH. This new 5-carbon molecule is called ribulose-5-phosphate.

what is the upstream fermentation:

In upstream bioprocessing, microbes are screened, cultured, and grown inside a bioreactor, providing necessary nutrients and growth conditions.

write the co-enzyme used by transketolase:

THIAMINE PYROPHOSPHATE, A COENZYME OF TRANSKETOLASE.

Phases of Batch fermentation?

(i) lag phase (adapt to their surroundings)

(ii) exponential growth (grow in numbers)

(iii) stationary phase (stop growing)

(iv) death phase

Pentose Phosphate Pathway?

The pentose phosphate pathway is a metabolic pathway parallel to glycolysis. It generates NADPH and pentoses as well as ribose 5-phosphate, a precursor for the synthesize of nucleotides. While the pentose phosphate pathway does involve oxidation of glucose, its primary role is anabolic rather than catabolic.

The mammalian ribosome contains two major nucleoprotein subunits:

- A larger one with 60S
- A smaller one with 40S.

NAD+ (nicotinamide adenine dinucleotide)

NADP+ (nicotinamide adenine dinucleotide phosphate)

Small nuclear RNA (snRNA)- play pivotal roles in RNA processing, particularly mRNA processing

Types of RNA?

(i) **Messenger RNAs (mRNAs)**- transfer genetic information from DNA to the protein synthesizing machinery.

(ii) **Ribosomal RNAs (rRNAs)**- contribute to the formation and function of ribosomes

(iii) **Transfer RNAs (tRNAs)-** adapter molecules that carry specific amino acids for protein synthesis

(iv) **Small nuclear RNA (snRNA)-** play pivotal roles in RNA processing, particularly mRNA processing

(v) **Ribozymes** — some RNA molecules have intrinsic catalytic activity these RNA enzymes, are called ribozymes.

The enzyme used in Glycolysis?

(i) Hexokinase

(ii) Phosphofructokinase (iii) Pyruvate kinase.

Elimination?

Loss of water from an alcohol introduction of a C=C bond. The loss of water from an alcohol, resulting in the introduction of a C=C bond b/w 2 adjacent carbon atoms.

NcRNA:

A non-coding RNA (ncRNA) is an RNA molecule that is not translated into a protein.